Master's Thesis Research Direction

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> *The familiar probability function or functional...provide mere continuum idealizations and by reason of this circumstance conceal the information-theoretic source from which they derive....Every physical quantity, every it, derives its ultimate significance from bits, binary yes-or-no indications, a conclusion which we epitomize in the phrase,* it from bit.
>
> —John Archibald Wheeler

The learning problem begins with *time*, or the lack thereof. Because both humans and machines have a finite amount of time to interact with the world and perform computation, what is physically even possible to learn is also limited. What is the fundamental unit of time? For a learner, it will be the *bit*—each unit of time may yield just one more bit of information, whether through observation, interaction, or computation.

This fundamental limitation of learning is fortunate for us—otherwise, we may never learn how learning works. In fact, beginning with the bit may suggest a learning framework that is general yet structured. And we may build on existing knowledge of:

1. algebra and topology
2. information theory
3. complexity theory
4. functional analysis
5. category theory

Let's briefly expand on these connections.

Summary: Information can be represented topologically—we may compute a function $f : X \to Y$ with respect to our knowledge if and only if it is continuous under the 'right' topology. Intuitively, an open set $U \subset X$ encodes knowledge that points in U are distinguishable from points outside of U.[1]

For example, at first glance, the two pandas below are topologically indistinguishable to us, for we humans don't have pixel-level information. So we classify both the same way.

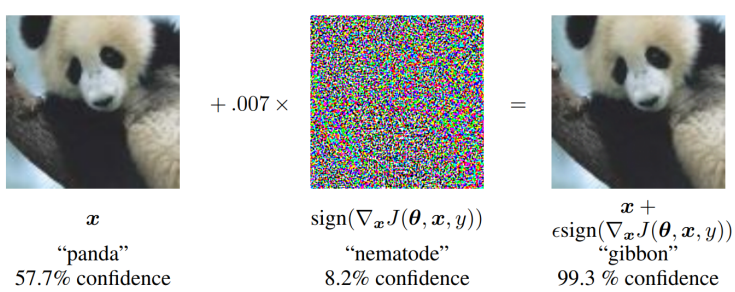

Figure 1: Famous adversarial example on GoogLeNet.[G2014]

[1] Technically, the interior of the complement.

However, the GoogLeNet classifier has access to more information, allowing it to (undesirably) distinguish between these two images. To us, the classifier appears discontinuous or random, as it seems to arbitrarily classify the same image in different ways.

The notion of *gaining information* then corresponds to refining a topology. A finer topology enables a more nuanced view of the world (e.g. an equestrian might easily identify the breed, age, and sex of a horse, but to me, horses just look like horses). On the other hand, there are tradeoffs of time and complexity.

This view is able to generalize:

- communication protocols: Alice and Bob are jointly trying to compute a function $f : X \times Y \to Z$ where Alice has input x and Bob has input y. What is the least number of bits of communication needed so that both may compute $f(x, y)$? In this language, we say that Alice and Bob initially have open sets corresponding with cylinder sets; for example, Alice has open sets of the form $\{x\} \times Y$. Communication refines their respective topologies. Here, what is the least number of bits of communication needed so that f is continuous for both Alice and Bob?

  In fact, [K2017] sets up a framework in which learning may be seen as a communication protocol (communication between learner and world/teacher, say). And from there, we can draw connections to complexity theory as well.

- hidden Markov models: the states of a HMM can be seen as open sets partitioning a space X. The hidden action on the space $f : X \to X$ is seen through the lens of the partition, $f : X/_\sim \to X/_\sim$; however, f is discontinuous on the coarser topology (it doesn't respect the partition) and so f appears to be a random function.

- generalization error: if the goal is to learn a classification $h^* : X \to Y$, then the learner should also determine the coarsest topology on which h^* is still continuous (and thus forcing the learned classifier to be regular, so helping to rule out adversarial examples). The collection of possible topologies on X is a poset (from the coarsest topology—the trivial topology—to the finest possible—which is not necessarily the discrete topology). We want the coarsest topology for which h^* is still continuous, as a finer topology implies using more information/computation. In general, the fineness of a topology corresponds to how much information is needed to learn (and may be unbounded).

Mode of Thought: It should be emphasized that the topological structure on X arises from considering collections of yes-or-no questions $f : X \to \{0, 1\}$ on X (in learning theory, we might think of f as a feature or query). That is, Boolean functions are first-class citizens in this framework; they *induce* a topology on X.

This perspective is quite similar to many *kernel* methods. There, X is also arbitrary. But so long as we have a function $f : X \times X \to \mathbb{R}$ satisfying some reasonable conditions we require of a 'similarity measure', then there is a natural way to view X as points inside a Hilbert space, thus informing X with structure.

Philosophically, this mode of thinking draws inspiration from the observer-world relationship from quantum mechanics—functions as observation, interaction, or computation. As Wheeler writes, "all things physical are information-theoretic in origin and this is a participatory universe." [W1990]

Example Research Question: Consider the problem of overfitting and adversarial examples, as in Figure 1. The setup is this: a learning model is trained on a relatively small collection of samples $S \subset X \times Y$, to produce a classifier $f : X \to Y$.

The problem is that we naively treat each training point $p = (x, y)$ as a point (i.e. an infinitely precise object, modulo the physical capabilities of a computer). However, p is actually meant to be a *representative* of a collection of equivalent images (i.e. the open set of points that are topologically indistinguishable from p). And so, p is *not* an infinitely precise object, and so we must *propagate errors* (c.f. significant figures).

In neural networks, one consequence is that the calculated loss function is discontinuous with respect to the desired topology—it represents the loss if we consider the training samples essentially as infinitely precise objects. Perhaps one way we can overcome this is by smoothing the loss function (a Gaussian blur, say). One possible avenue of research in addition to solidifying this heuristic theoretically is to see if there is a computationally efficient way to do this.

Readings

The following is a list of papers and books that, based on a cursory preview, seems important to understand to pursue learning theory further in the direction described. Probably not all will be relevant, and most likely the list will continue to change over time.

Philosophical and Mathematical Foundations (4)

- [W1990] *Information, physics, quantum: the search for links.* Essay by the famous physicist, John Archibald Wheeler, on the information-theoretic basis of existence.
- [F2013] *Where the 'it from bit' comes from.* Comments on Wheeler's essay.
- [N2011] *Classical and quantum mechanics via Lie algebras.* "This book presents classical mechanics, quantum mechanics, and statistical mechanics in an almost completely algebraic setting." (See also chapters on information theory).
- [O2005] *Prevalence.* "An analogue of the finite-dimensional notions of 'Lebesgue almost every' and 'Lebesgue measure zero' in the infinite-dimensional setting. The theory of prevalence...provides a powerful framework for describing generic behavior in a probabilistic way."

Algebras and Boolean Functions (4)

Keywords: lattices, Heyting algebra

- [B2012] *A Course in Universal Algebra.* (Foundations)
- [W1987] *On the lattice of Fréchet-Nikodým topologies.* (Foundations)
- [K1994] *Cryptographic limitations on learning Boolean formulae and finite automata.* "We prove the intractability of learning several classes of Boolean functions in the distribution-free model (also called the Probably Approximately Correct or PAC model) of learning from examples."
- [M2016] *Shattered sets and the Hilbert function.* "We study complexity measures on subsets of the boolean hypercube and exhibit connections between algebra (the Hilbert function) and combinatorics (VC theory). These connections yield results in both directions."

Generalization Theory (6)

Keywords: information bottleneck

- [S2017] *Opening the black box of deep neural networks via information.* "Most of the training epochs in standard DL are spent on compression of the input to efficient representation and not on fitting the training labels...The converged layers lie on or very close to the Information Bottleneck (IB) theoretical bound."
- [S2016] *The deterministic information bottleneck.* (Foundations)
- [T2000] *The information bottleneck method.* (Foundations)
- [V2002] *Learning and Generalization.* (Foundations) Note list of open problems at the end.
- [P2017] *Why and when can deep—but not shallow—networks avoid the curse of dimensionality?* "Deep networks have the theoretical guarantee, which shallow networks do not have, that they can avoid the curse of dimensionality for an important class of problems, corresponding to compositional functions, that is functions of functions."
- [Z2017] *mixup: Beyond empirical risk minimization.* "We propose *mixup*, a simple learning principle to alleviate [memorization and sensitivity to adversarial examples]."

Information Theory (6)

Keywords: communication complexity

- [K2017] *On communication complexity of classification problems.* Introduces communication protocol framework for distributed learning. Connections to P and NP classes.
- [J2017] *Multivariate dependence beyond Shannon information.* "The vast majority of...informational measures are simply inadequate for determining the meaningful dependency structure within joint probability distributions."
- [M2010] *Information Theory and Statistical Physics—Lecture Notes.* Connections between information theory and statistical physics.
- [C2012] *Elements of Information Theory.* (Foundations)
- [B2014] *Information equals amortized communication.* (Foundations)
- [Y1997] *Assouad, Fano, and Le Cam.* (Bounding techniques) Three commonly used methods for constructing minimax lower bounds.

Learning Theory (10)

Keywords: queries, bounds

- [A1988] *Queries and concept learning.* (Foundations)
- [A2014] *Provable bounds for learning some deep representations.* "Algorithms with provable guarantees that learn a class of deep nets in the generative model view popularized by Hinton and others...the analysis of the algorithm reveals interesting structure of neural networks with random edge weights."
- [B2004] *Limits on exact learning from membership and equivalence queries.* "Different combinatorial properties of concept classes that give bounds on the query complexity of exact learning."
- [F2009] *On the power of membership queries in agnostic learning.* "We give a simple proof that any concept class learnable agnostically by a distribution-independent algorithm with access to membership queries is also learnable agnostically without membership queries."
- [H2015] *Minimax Analysis of Active Learning.* "This work establishes distribution-free upper and lower bounds on the minimax label complexity of active learning with general hypothesis classes, under various noise models."
- [K1993] *Active learning using arbitrary binary valued queries.* "Bounds on the improvement in sample complexity that can be expected from using oracles, we consider active learning in the sense that the learner has complete control over the information received."
- [K1996] *Learning and Geometry: Computational Approaches.* (Chapter 3: A Review of Some Extensions to the PAC Learning Model).
- [R2011] *Lower bounds for passive and active learning.* "We develop unified information-theoretic machinery for deriving lower bounds for passive and active learning schemes. Our bounds involve the so-called Alexander's capacity function."
- [W2015] *A compression technique for analyzing disagreement-based active learning.* "We introduce a new and improved characterization of the label complexity of disagreement-based active learning, in which the leading quantity is the *version space compression set size*."
- [Y2010] *Bayesian active learning using arbitrary binary valued queries.* "[In] a general Bayesian active learning setting, in which the learner can ask arbitrary yes/no questions...we derive upper and lower bounds."

Neural Networks (8)

Keywords: spin-glass

- [C2015] *The loss surfaces of multilayer networks.* "We study the connection between the highly non-convex loss function of a simple model of the fully-connected feed-forward neural network and the Hamiltonian of the spherical spin-glass model."
- [G2016] *A theoretically grounded application of dropout in recurrent neural networks.* "Grounding of dropout in approximate Bayesian inference suggests an extension of the theoretical results, offering insights into the use of dropout with RNN models. We apply this new variational inference based dropout technique...outperform[ing] existing techniques."
- [K'2017] *Overcoming catastrophic forgetting in neural networks.* "Neural networks are not, in general, capable of [learning sequentially] and it has been widely thought that catastrophic forgetting is an inevitable feature of connectionist models. We show that it is possible to overcome this limitation."
- [M2014] *An exact mapping between the Variational Renormalization Group and Deep Learning.* "We show that deep learning is intimately related to one of the most important and successful techniques in theoretical physics, the renormalization group. RG is an iterative coarse-graining scheme that allows for the extraction of relevant features (i.e. operators)."
- [P2007] *Characterizing the function space for Bayesian kernel models.* "We study a coherent Bayesian kernel model based on an integral operator defined as the convolution of a kernel with a signed measure. Priors on the random signed measures correspond to prior distributions on the functions mapped by the integral operator."
- [S'2017] *Randomness in neural networks: an overview.* "Randomly assign a subset of the networks' weights so that the resulting optimization task can be formulated as a linear least-squares problem. This methodology can be applied to both feedforward and recurrent networks, and similar techniques can be used to approximate kernel functions."
- [S2015] *Deep learning in neural networks: An overview.* (Foundations)
- [W2013] *Topological and dynamical complexity of random neural networks.* "Random neural networks are dynamical descriptions of randomly interconnected neural units. These show a phase transition to chaos as a disorder parameter is increased. The microscopic mechanisms underlying this phase transition are unknown, and similarly to spin-glasses, shall be fundamentally related to the behavior of the system."

Operator Theory and Kernels (9)

Keywords:

- [A1950] *Theory of reproducing kernels.* (Foundations)
- [P2016] *An introduction to the theory of reproducing kernel Hilbert spaces.* (Foundations)
- [F2016] *Bayesian learning of kernel embeddings.* "We propose a new probabilistic model for kernel mean embeddings...combining a Gaussian process prior over the RKHS containing the mean embedding with a conjugate likelihood function, thus yielding a closed form posterior over the mean embedding."
- [M†2016] *Conditionally positive definite kernels in Hilbert C^*-modules.* "Characterization of the conditionally positive definiteness." (Need to understand the GNS construction first).
- [M2006] *Mercer's theorem, feature maps, and smoothing.* (Foundations) "We study Mercer's theorem and feature maps for several positive definite kernels that are widely used in practice."

- [M†2006] *Universal kernels.* Conditions for a universal kernel.
- [S2001] *The kernel trick for distances.* (Foundations) Generalize kernel methods to distances.
- [V2016] *An operator theoretic approach to nonparametric mixture models.* "we introduce general spectral algorithms for recovering the mixture components and illustrate their use on a synthetic data set"
- [S1976] *Positive definite functions and generalizations, an historical survey.* (Foundations)

Probability Theory (2)

Keywords: projective limits (c.f. category theory)

- [R1971] *Projective limits of probability spaces.* Note: Cantor space may be a projective limit...
- [R†2011] *Foundations of Stochastic Analysis.* "Fundamental theories and contrasts the natural interplay between real and abstract methods."

Abstract Nonsense and More Math (8)

Keywords: Polish spaces, Baire spaces, frames, locales, sheaves, Stone duality, uniform spaces, entourages, gauge spaces, topos theory

- [G2013] *Aspects of classical descriptive set theory.* (Foundations) Applications to functional analysis, ergodic theory, operator algebras, group action, logic.
- [J1983] *The point of pointless topology.* (Foundations)
- [E2004] *Synthetic topology: of data types and classical spaces.* (Foundations) Equivalence of computability and continuity.
- [W2016] *Preliminaries on Dynamical Systems and Stability Theory.* (Foundations)
- [B2010] *Functional analysis, Sobolev spaces and partial differential equations.* (Foundations)
- [M1999] *Algebra.* (Foundations)
- [M1971] *Category Theory for the Working Mathematician.* (Foundations)
- [M2012] *Sheaves in Geometry and Logic.* (Foundations)

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